

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

Sirui Lu

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Chapter 1

Introduction

Chapter 2

Matrix Product States

Fix a physical dimension $d \geq 1$ and a bond dimension $D \geq 1$. All MPS in this blueprint are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$.

Definition 2.2 (Word evaluation). Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{I}_D$.

Lemma 2.3 (Word evaluation respects concatenation). *For any words w_1, w_2 , $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.*

Proof. Induction on w_1 . □

Definition 2.4 (Trace coefficient). The *trace coefficient* of a word w is $c_w(A) := \text{tr}(A^w)$.

Definition 2.5 (Matrix product vector). The *matrix product vector* (MPV) of an MPS tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$.

Definition 2.6 (Transfer map). The *transfer map* associated to an MPS tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Lemma 2.7 (Transfer map is positive). *If $X \geq 0$, then $\mathcal{E}_A(X) \geq 0$.*

Proof. Each summand $A^i X (A^i)^\dagger$ is PSD when X is PSD. $\square$

2.2 Gauge equivalence and same MPV

Definition 2.8 (Gauge equivalence). Two MPS tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that

$$B^i = X A^i X^{-1} \quad \text{for all } i \in \{0, \dots, d-1\}.$$

Definition 2.9 (Same MPV). Two MPS tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size $N \geq 1$.

Definition 2.10 (Same MPV — different bond dimensions). For MPS tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N \geq 1$.

Definition 2.11 (Proportional MPV). Two MPS tensors A and B (of possibly different bond dimensions) generate *proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Lemma 2.12 (Word evaluation under scaling). *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w .

Proof. Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . $\square$

Definition 2.13 (Gauge-phase equivalence). Two MPS tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C}$ such that

$$B^i = \zeta X A^i X^{-1} \quad \text{for all } i.$$

Lemma 2.14 (Word evaluation under conjugation). *If $B^i = X A^i X^{-1}$ for all i , then for every word w ,*

$$B^w = X A^w X^{-1}.$$

Proof. Induction on the word length, using $XX^{-1} = \mathbb{I}$. $\square$

Theorem 2.15 (Gauge invariance of MPVs). *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. Let $B^i = X A^i X^{-1}$. By Lemma 2.14, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. $\square$

2.3 Injectivity and normality

Definition 2.16 (Injectivity). An MPS tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.17 (L -block injectivity). An MPS tensor A is *L -block injective* if $\text{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Definition 2.18 (Normal tensor). An MPS tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0 -block injective.

2.4 Canonical form

Definition 2.19 (Canonical form). A *canonical form* for an MPS consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated MPS tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This decomposition is defined in [PGVWC07] and reviewed in [CPGSV21].

Remark 2.20. The formalization requires each block tensor to be injective (Definition 2.16). More generally, canonical forms in the literature [CPGSV21] allow normal tensors (Definition 2.18), which become injective after sufficient blocking (Definition 2.23).

Definition 2.21 (Block-diagonal MPS tensor). Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the *block-diagonal MPS tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Theorem 2.22 (MPV decomposition). *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Proof. For each physical index i , the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.12, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.5 Blocking

Definition 2.23 (Blocked tensor). Given an MPS tensor A and a blocking length $L \geq 1$, the *L-blocked tensor* is the MPS tensor with physical dimension d^L and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Lemma 2.24 (Blocked word evaluation). *Let $w = (J_1, \dots, J_m)$ be a word in the blocked alphabet (each $J_k \in \{0, \dots, d^L-1\}$). Decoding each blocked index J_k back to an L -tuple of physical indices and concatenating gives a word $\tilde{w}$ of length mL in the original alphabet. Then $(A^{[L]})^w = A^{\tilde{w}}$.*

Proof. Induction on w , using the concatenation lemma (Lemma 2.3). $\square$

Lemma 2.25 (Blocking preserves MPV coefficients). *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system (with $\sigma_k \in \{0, \dots, d^L - 1\}$), there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. Flatten the blocked basis state and apply the blocked word evaluation lemma (Lemma 2.24). $\square$

Theorem 2.26 (Blocking preserves same MPV). *If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. By Lemma 2.25, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the flattened configuration. The same holds for $B^{[L]}$ (with the same flattened configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.6 MPV overlap

Definition 2.27 (MPV as Hilbert-space vector). We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_\sigma$ as in Definition 2.5. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.28 (MPV inner product). The *MPV inner product* between two MPS tensors A and B at system size N is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.29 (MPV overlap). The *MPV overlap* between two MPS tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Lemma 2.30 (Overlap equals conjugate inner product). $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. $\square$

Chapter 3

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B (of the same bond dimension), then A and B are gauge equivalent.

3.1 Trace pairing

Theorem 3.1 (Nondegeneracy of the trace pairing). *For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\text{tr}(E_{ji}M) = M_{ij}$. Hence if $\text{tr}(MN) = 0$ for all N , then every entry of M vanishes. $\square$

Lemma 3.2 (Same MPV implies trace agreement). *If A and B generate the same MPV family, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .*

Proof. Viewing the word w as a configuration of length $|w|$, the MPV coefficient at this configuration is exactly $\text{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. $\square$

Lemma 3.3 (Length-2 trace identity). *If A and B generate the same MPV family, then $\text{tr}(A^i A^j) = \text{tr}(B^i B^j)$ for all $i, j \in \{0, \dots, d-1\}$.*

Proof. Apply Lemma 3.2 to the word $w = (i, j)$. $\square$

Definition 3.4 (Trace pairing map). For an MPS tensor A , define the linear map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ by

$$\Phi_A(M)_i = \text{tr}(M \cdot A^i).$$

Theorem 3.5 (Injectivity of the trace pairing map). *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. If $\Phi_A(M) = 0$, then $\text{tr}(M \cdot A^i) = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\text{tr}(MN) = 0$ for all N , so $M = 0$ by Theorem 3.1. $\square$

Lemma 3.6 (Injectivity transfer via range inclusion). *If $\Phi_A, \Phi_B : V \rightarrow W$ are linear maps with $\ker \Phi_A = \{0\}$ and $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$, then $\ker \Phi_B = \{0\}$.*

Proof. By rank-nullity, $\ker \Phi_A = \{0\}$ gives $\dim(\text{range}(\Phi_A)) = \dim V$. The range inclusion forces $\dim(\text{range}(\Phi_B)) \geq \dim V$, so $\dim(\ker \Phi_B) = 0$ by rank-nullity again. $\square$

Theorem 3.7 (Nonvanishing trace on injective tensors). *If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i , then we have a contradiction.*

Proof. If $B^i = 0$ for all i , then Lemma 3.2 gives $\text{tr}(A^w) = 0$ for every word w , in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

3.2 Linear extension

Theorem 3.8 (Existence and uniqueness of the linear extension). *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i .*

Proof. Write $\ell_A : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). By the length-2 trace identity (Lemma 3.3), $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$. By Theorem 3.5, $\ker \Phi_A = \{0\}$; Lemma 3.6 then forces $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 3.9 (Multiplicativity of the linear extension). *The linear extension T from Theorem 3.8 is multiplicative: $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. **Step 1** ($\Phi_B \circ T = \Phi_A$). By the length-2 trace identity (Lemma 3.3), $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). The length-3 trace identity gives $\text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k)$ for all i, j, k . So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. Injectivity of Φ_B (from Lemma 3.6) gives $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j , the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

3.3 Simplicity and Skolem–Noether

Theorem 3.10 (Simplicity of $M_D(\mathbb{C})$). *A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 3.11 (Skolem–Noether). *Every $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \text{Aut}_{\mathbb{C}\text{-alg}}(M_D(\mathbb{C}))$, there exists $X \in \text{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M .*

Proof. Via the canonical identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\text{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X . $\square$

Lemma 3.12 (Promotion to algebra homomorphism). *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$, and T is a $\mathbb{C}$ -algebra homomorphism.*

Proof. By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$ -algebra homomorphism. $\square$

3.4 Assembly

Theorem 3.13 (Single-block Fundamental Theorem). *Let A be an injective MPS tensor and let B be an MPS tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i .*

Proof. By Theorem 3.8, there is a unique linear T with $T(A^i) = B^i$. By Theorem 3.9, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i , contradicting Theorem 3.7. By Theorem 3.10, T is bijective. By Lemma 3.12, T is a $\mathbb{C}$ -algebra automorphism. Theorem 3.11 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$.

Definition 4.2 (Trace-preserving map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Definition 4.3 (Quantum channel). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both positive and trace-preserving. In [Wol12], “channel” refers to completely positive trace-preserving maps; here we only require positivity.

Definition 4.4 (Density matrices). The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 4.5 (Density matrices are compact). $\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).

Proof. The PSD cone is closed (preimage of the closed nonneg cone under continuous quadratic forms), and PSD matrices with $\text{tr}(\rho) = 1$ have entry norms bounded by 1 (diagonal entries by trace; off-diagonal by Cauchy–Schwarz). Heine–Borel gives compactness. $\square$

Theorem 4.6 (Density matrices are convex). $\mathcal{D}_D$ is convex.

Proof. Convex combination of PSD matrices is PSD, and trace is linear. $\square$

Theorem 4.7 (Density matrices are nonempty). For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem 4.8 (Channels preserve density matrices). If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. Positivity preserves $\rho \geq 0$, and trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

4.2 Kadison–Schwarz inequality

Definition 4.9 (Kraus map). Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$.

Definition 4.10 (Adjoint Kraus map). The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A , the adjoint Kraus map corresponds to the transfer map $\mathcal{E}_A$ with the roles of A^i and $(A^i)^\dagger$ exchanged.

Definition 4.11 (Unital Kraus map). A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{I}$.

Definition 4.12 (Trace-preserving Kraus map). A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{I}$. This is the standard MPS normalization condition.

Theorem 4.13 (Kadison–Schwarz inequality). Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{I}$). Then for all $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. The 2×2 block matrix $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{I} \end{bmatrix}$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{I}]^T$). Applying the unital CP map blockwise preserves PSD-ness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$ -block maps to $\mathbb{I}$). The Schur complement of the $(2,2)$ -block $\mathbb{I}$ gives the result. $\square$

Theorem 4.14 (Kadison–Schwarz for the adjoint channel). If $\sum_i K_i^\dagger K_i = \mathbb{I}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.

Proof. The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{I}$, so they form a unital family. Apply Theorem 4.13 to $\{K_i^\dagger\}$. $\square$

Theorem 4.15 (Hilbert–Schmidt contraction). If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.

Proof. By Theorem 4.13, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

4.3 Multiplicative domain

Theorem 4.16 (KS gap decomposition). For a unital Kraus map E , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := X K_i^\dagger - K_i^\dagger E(X)$.

Proof. Expand the right-hand side $\sum_i (X K_i^\dagger - K_i^\dagger E(X))^\dagger (X K_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{I}$ to identify the result with the KS gap. $\square$

Theorem 4.17 (KS equality implies Kraus commutation). Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then $X K_i^\dagger = K_i^\dagger E(X)$ for all i .

Proof. By Theorem 4.16, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. $\square$

Theorem 4.18 (KS equality for peripheral eigenvectors). *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.*

Proof. The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 4.13. Its trace satisfies $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$. So $\text{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. $\square$

Theorem 4.19 (Multiplicative domain). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. From Theorem 4.17, $XK_i^\dagger = K_i^\dagger E(X)$ for all i . Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \square \end{aligned}$$

4.4 Irreducibility

Definition 4.20 (Irreducible CP map). A completely positive linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{I}$. This is [Wol12, Thm. 6.6, condition (1)].

Theorem 4.21 (Injectivity implies irreducibility). *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{I} - P)A^i P = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{I} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{I}$. $\square$

Theorem 4.22 (Transfer map is a channel). *If $\sum_i (A^i)^\dagger A^i = \mathbb{I}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel.*

Proof. Positivity: each summand $A^i X (A^i)^\dagger \geq 0$ when $X \geq 0$. Trace preservation: $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

4.5 Cesàro fixed-point theorem

Definition 4.23 (Cesàro mean). The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.24 (Cesàro telescope). $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. $\square$

Theorem 4.25 (Hermitian fixed-point decomposition). *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_{\pm} \geq 0$, $\text{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

Proof. Let Q be the support projection of P_+ . Trace preservation and positivity give $\text{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_{\pm}) = P_{\pm}$. $\square$

Theorem 4.26 (Cesàro fixed-point theorem). *Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.7). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 4.5), extract a convergent subsequence $\sigma_{N_k} \rightarrow \rho$. The telescope identity (Theorem 4.24) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.13], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead, as Brouwer's theorem is not available in Mathlib. $\square$

4.6 Fixed-point projection

Definition 4.27 (Fixed-point projection). Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional.

Theorem 4.28 (Power decomposition). *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. P is idempotent ($P^2 = P$), and trace preservation gives $E \circ P = P$ and $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. $\square$

4.7 Peripheral spectrum and primitivity

Definition 4.29 (Peripheral spectrum). The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $\|\mu\|$ equal to the spectral radius of T .

Definition 4.30 (Peripheral eigenvalues). The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are those eigenvalues λ with $|\lambda|$ equal to the spectral radius of E . For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these are the eigenvalues on the unit circle.

Theorem 4.31 (1 is a peripheral eigenvalue). *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then $1 \in \text{peripheral}(E)$.*

Proof. ρ is an eigenvector with eigenvalue 1, and $|1| = 1$ equals the spectral radius. $\square$

Theorem 4.32 (Peripheral eigenvalues lie on the unit circle). *Every peripheral eigenvalue λ satisfies $|\lambda| = 1$.*

Proof. By definition of peripheral eigenvalues (spectral radius = 1 for channels). $\square$

Theorem 4.33 (Finiteness of peripheral eigenvalues). *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. $\square$

Theorem 4.34 (Peripheral eigenvalues are roots of unity). *Every peripheral eigenvalue of a channel is a root of unity. This follows from [Wol12, Prop. 6.2]: peripheral eigenvalues have trivial Jordan blocks, so their powers form a finite set.*

Proof. A complex number of modulus 1 whose powers take only finitely many values must be a root of unity. $\square$

Definition 4.35 (Channel period). The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\text{peripheral}(E)|$.

Definition 4.36 (Primitive channel). A channel is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. This corresponds to [Wol12, Thm. 6.11].

Theorem 4.37 (Primitivity equals period one). *A channel with a nonzero fixed point is primitive if and only if its period is 1.*

Proof. If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 4.31), the only element is 1. $\square$

Theorem 4.38 (Non-peripheral eigenvalues of primitive channels). *If a channel is primitive, then every eigenvalue $\lambda \neq 1$ satisfies $|\lambda| < 1$.*

Proof. By primitivity, no eigenvalue other than 1 has modulus 1. Since the spectral radius is 1, all other eigenvalues have strictly smaller modulus. $\square$

Chapter 5

Quantum Perron–Frobenius Theory

This chapter establishes that the transfer map of an injective MPS tensor has a unique positive semidefinite fixed point (up to scaling), and that this fixed point is positive definite. The results follow [EHK78] and [Wol12, Ch. 6].

5.1 Positive definiteness

Theorem 5.1 (PSD fixed point is positive definite). *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. Suppose $\rho v = 0$ for some $v \neq 0$. The fixed-point equation $\mathcal{E}_A(\rho) = \rho$ and the expansion $\mathcal{E}_A(\rho) = \sum_i A^i \rho (A^i)^\dagger$ give

$$v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v).$$

Each term is nonneg; if the sum is zero, then $\rho(A^i)^\dagger v = 0$ for all i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, this gives $\rho M v = 0$ for all M , hence $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.8], positive definiteness is proved using the growth condition $(\mathcal{K} + \mathcal{E})^{D-1}(X) > 0$ for irreducible maps ([Wol12, Thm. 6.6, condition (2)]). Our proof instead exploits injectivity of the MPS tensor directly, which avoids the irreducibility growth chain and gives a shorter argument. $\square$

5.2 Uniqueness

Theorem 5.2 (Uniqueness of PSD fixed point). *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. Both ρ and σ are positive definite by Theorem 5.1. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1} \sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H . Then $\tau = \sigma - c_0 \rho$ is a PSD fixed point of $\mathcal{E}_A$ which is *not* positive definite (it has a zero eigenvalue by construction). By Theorem 5.1, τ must be zero, giving $\sigma = c_0 \rho$. In [Wol12, Thm. 6.8], uniqueness follows from non-degeneracy of the spectral radius eigenvalue, proved via the $(\mathcal{K} + \mathcal{E})^{D-1}$ growth condition. Our proof reduces uniqueness to the positive definiteness theorem (Theorem 5.1) via a spectral shift argument, which is more elementary. $\square$

5.3 Existence and assembly

Theorem 5.3 (Existence of PSD fixed point). *Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{K}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{K}$, the transfer map is a quantum channel (Theorem 4.22). The Cesàro fixed-point theorem (Theorem 4.26) then provides the fixed point. $\square$

Theorem 5.4 (Quantum Perron–Frobenius). *Let A be an injective MPS tensor (with normalized Kraus operators). Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. Combine existence (Theorem 5.3), positive definiteness (Theorem 5.1), and uniqueness (Theorem 5.2). $\square$

5.4 Doubly stochastic gauge

Theorem 5.5 (Unital gauge). *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i (A'^i)^\dagger = \mathbb{K}$, i.e. the gauged Kraus map is unital.*

Proof. $\sum_i (S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}(\sum_i A^i \rho (A^i)^\dagger) (S^\dagger)^{-1} = S^{-1} \rho (S^\dagger)^{-1} = S^{-1} SS^\dagger (S^\dagger)^{-1} = \mathbb{K}$. $\square$

Theorem 5.6 (Trace-preserving gauge). *Let σ be a fixed point of the adjoint transfer map $\mathcal{E}_A^\dagger(\sigma) = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{K}$, i.e. the gauged Kraus map is trace-preserving.*

Proof. $\sum_i (SA^iS^{-1})^\dagger (SA^iS^{-1}) = (S^\dagger)^{-1}(\sum_i (A^i)^\dagger \sigma A^i) S^{-1} = (S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{K}$. $\square$

Chapter 6

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D , the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

6.1 Cumulative span

Definition 6.1 (Word span). The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_n} : (i_1, \dots, i_n) \in \{0, \dots, d-1\}^n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Eq. (1)].

Definition 6.2 (Cumulative span). The *cumulative span* at level n is

$$T_n(A) = \text{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_m} : m \leq n\} = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Theorem 6.3 (Cumulative span stabilisation). *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.*

Proof. If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$, so left-multiplying by any A^i gives $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m \geq n+1$, hence $T_m = T_n$. $\square$

Theorem 6.4 (Dimension growth). *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. $T_n \subseteq T_{n+1}$ and $T_n \neq T_{n+1}$ imply strict rank increase. $\square$

6.2 Fitting decomposition

Definition 6.5 (Fitting decomposition). A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,

4. the generalized eigenspaces are linearly independent.

Theorem 6.6 (Fitting decomposition exists). *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard spectral theory (available in Mathlib). In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. Our formalization uses the Fitting decomposition (generalized eigenspaces) as a substitute, since the full Jordan normal form is not available in Mathlib. $\square$

6.3 Nonzero trace product

Theorem 6.7 (Cumulative span reaches $M_D(\mathbb{C})$). *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 6.4, if $T_n \neq T_{n+1}$ then the dimension increases by ≥ 1 . Since $\dim M_D(\mathbb{C}) = D^2$, after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 , hence $T_{L_0} = M_D(\mathbb{C})$). The proof follows [SPGWC10, Lemma 1] and [Wol12, Lemma 6.2]. Those references obtain the sharper bound $D^2 - d + 1$ by starting the dimension count at $\dim S_1 = d$; our formalization uses the coarser bound D^2 . $\square$

Theorem 6.8 (Nonzero trace product). *If A is normal, then there exists a word w of length $\leq D^2$ such that $\text{tr}(A^{i_1} \cdots A^{i_{|w|}}) \neq 0$. This is [SPGWC10, Lemma 1].*

Proof. By Theorem 6.7, the identity $\mathbb{I} \in T_{D^2}(A)$, so $\mathbb{I}$ is a linear combination of word products of length $\leq D^2$. At least one of these word products must have nonzero trace (since $\text{tr}(\mathbb{I}) = D \neq 0$). $\square$

6.4 Eigenvector spreading

Definition 6.9 (Cumulative vector span). Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_m} \varphi : m \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 6.10 (Eigenvector spreading). *Let A be a normal MPS tensor and let $A^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises. But $T_N(A) = M_D(\mathbb{C})$ by Theorem 6.7, so every standard basis matrix is in the cumulative span; applying these to $\varphi \neq 0$ gives all of $\mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. $\square$

6.5 Rank-one products and eigenvalue extraction

Theorem 6.11 (Eigenvalue extraction). *If A is normal, then there exist a word w , a nonzero scalar μ , and a nonzero vector φ such that $(\text{evalWord } A \ w)\varphi = \mu\varphi$.*

Proof. By Theorem 6.8, there is a word w with $\text{tr}(\text{evalWord } A \ w) \neq 0$. A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues). The Fitting decomposition (Theorem 6.6) shows that the nonzero generalized eigenspace is nontrivial, yielding the eigenvector. In [SPGWC10] and [Wol12], this step uses the Jordan normal form directly; our formalization uses the Fitting decomposition instead, as the full Jordan normal form is not available in Mathlib. This suffices because we only need the existence of an eigenvector, not the full Jordan block structure. $\square$

6.6 Assembly

Definition 6.12 (Wielandt analysis). A *Wielandt analysis* for a normal MPS tensor A packages: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ , and a nonzero eigenvector φ of the matrix A^{w_0} , such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 6.13 (Wielandt analysis exists). *Every normal MPS tensor admits a Wielandt analysis.*

Proof. Immediate from Theorem 6.11. $\square$

Theorem 6.14 (Wielandt bound). *If A is a normal MPS tensor with bond dimension D , then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2 . This is [SPGWC10, Theorem 1] (which gives the sharper bound $D^2 - d + 1$ in certain cases; our formalization proves the D^2 bound).*

Proof. Immediate from Theorem 6.7. $\square$

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